

AI61201: Visual Computing with AI/ML (Autumn 2026)

Programming Assignment 1: Working With Images Using OpenCV (20 Marks)

Instructions: Complete all the tasks given, keeping the following points in mind:

- Implement the image operations from scratch (without using the OpenCV library) and verify using OpenCV library functions.
- Vectorized implementations are possible for most operations, and using for or while loops to access individual pixels is strongly discouraged due to its inefficiency. You can use NumPy to perform the mathematical operations.

Task 1: Loading and retrieving RGB channels of an image (1 Mark)

- Load iMAgEa.jpg ( Assignment-1 resources) using OpenCV and display it in the RGB space.
- Display its individual color channels (R, G, B) with proper labels.

Task 2: Color Space Conversion (6 Marks)

Task 2.1: YCbCr Operations (3 Marks)

- Convert the iMAgEa.jpg into **the YCbCr color space**, and display the Y, Cb, and Cr channels individually.
- Increase the brightness of the image by a constant value using the YCbCr image (*Note: clip brightness levels to the range of [0, 255]*).
- Convert the image back to RGB and display it. Compare your results with those obtained by using OpenCV functions.

Task 2.2: HSV Operations (3 Marks)

- Convert the iMAgEa.jpg to **HSV color space**.
- Increase the saturation of all colors to the maximum values.
- Convert back to the RGB color space and show the result. Compare your results with those obtained by using OpenCV functions.

Task 3: Bit Plane Operations (4 Marks)

- Load iIMAGEb.jpg ([Assignment-1 resources](#)). Convert the image to greyscale and toggle alternate bit planes of the image, i.e., keep bit planes 8, 6, 4, and 2 as they are and toggle bit planes 7, 5, 3, and 1.
- Display the greyscale image with toggled bit planes.

Task 4: Spatial Transformations and Interpolation (9 Marks)

Load iIMAGEc.jpg ([Assignment-1 resources](#)).

Task 4.1: Cropping and Rotation (4 Marks)

- Crop the central patch of size 300x200.
- Rotate the cropped patch by 45 degrees.

Task 4.2: Resizing and Interpolation (5 Marks)

- Resize the patch to upsample it by a factor of 2 using Lanczos interpolation. Compare your results with those obtained by using OpenCV functions.

Lanczos kernel:

$$L(x) = \begin{cases} 1 & \text{if } x = 0, \\ \frac{a \sin(\pi x) \sin(\pi x/a)}{\pi^2 x^2} & \text{if } -a \leq x < a \text{ and } x \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Link to download the inputs for all tasks:

[Assignment 1 Resources](#)

Submission Guidelines

1. The content that you submit must be your individual work.
2. Submit your code in .py as well as in .ipynb file formats. Both these file submissions are required to receive credit for this assignment
3. Ensure your code is well-commented and easy to follow. You can write your answers and explanations using text cells in the Jupyter Notebook files wherever required.
4. Submit output files wherever instructed.

5. The files should be named as “<roll_number>_assignment_1”. For example, if your roll number is 23AI91R01, the code the required file names will be 23AI91R01_assignment_1.py and 23AI91R01_assignment_1.ipynb. You should place all these files within a single zip file and upload it to Moodle as 23AI91R01_assignment_1.zip. Do not use .rar files for submission.
6. All submissions must be made through Moodle before the deadline. The submission portal will close at the specified time, and submissions via email will not be accepted.

TA for this assignment: Chandra Sekhar Lokesh Manem

If you have any queries regarding Assignment 1, please email
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